

hal 1

Langkah 1 : XOR utk a, x0, c

$J = 79051201001010$

0: 1112 = 0110111

$$r = 114 \Rightarrow 01110010$$

$$a = J \oplus 0 = 01001010 \oplus 01101111 = 00100101 = 37$$

$$x_0 : 0 \oplus r = 0101111 \oplus 01110010 : 00011101 = 29$$

$C = J \oplus r : 01001010 \oplus 01110010 = 00111000 : 56$

$$x_{n+1} = (a \cdot x_n + c) \bmod m$$

20. State pertama $(x_1, \dots, x_{20})$, dgn $x_0 = 29$;

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
x	2	15	13	8	7	16	9	20	14	22	19	0	10	12	17	18	9	21	5	11

- Channel ($x \bmod 7 \rightarrow \bmod 3$) Vite 15B = 2, 1, 6, 10, 2, 4, 6, 0, 1, 5, 0, 3, 5, 3, 9, 2, 0, 5, 4,

$$\text{mod } 3 \rightarrow R, G, G, R, R, G, R, R, G, R(?)$$

- Jumlah brt Sisip (C mod $a+1$): 3, 9, 2, 1, 9, 1, 1, 1, 3, 3, 4, 1, 3, 1, 2, 3, 2, 2, 4.

Cangkuk 3 : Pesan

Left (Jordanet, 5) kref home: "JORDA"

$J = 01001010$ $O = 01001111$ $F = 01000101$ $D = 01000100$

A: 01000001

bit stream (to bit) : 01001010011110101001010001001000001

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Catatan 4: Cover (maka 5x5 nilai bit, unit)

Pixel	R	G	B
P0	50	80	120
P5	60	87	133
P10	70	99	146
P15	80	101	159
P20	90	108	172

Catatan 5: Tabel Proses penggantian (LSB Replacement)

~~Pixel~~ ~~Channel~~ ~~Jlh bit~~ ~~Bit sisip~~ ~~Nilai lama~~ ~~Biner lama~~ ~~Nilai baru~~ ~~Biner baru~~

	Pixel	Channel	Jlh bit	Bit sisip	Nilai lama	Biner lama	Nilai baru	Biner baru
1	P2	B	3	010	130	1000010	130	1000010
2	P5	G	4	0101	101	01100101	101	01100101
3	P15	R	2	00	79	01001111	79	01001100
4	P8	G	1	1	120	01111000	121	01111001
5	P7	R	4	0011	66	01000010	67	01000011
6	P16	B	1	1	164	10100100	165	10100101
7	P9	G	1	1	124	01111000	125	01111001
8	P20	R	1	0	90	01011010	90	01011010
9	P14	R	3	101	87	01010010	85	01010001
10	P22	G	3	001	150	10000010	129	10000001
11	P19	B	4	0010	179	10110011	178	10110010
12	P0	R	1	0	50	00110010	50	00110010
13	P10	R	3	010	70	01000110	66	01000100
14	P12	B	1	0	156	10011100	156	10011100
15	P17	R	2	01	86	01010110	85	01010101
16	P13	G	3	000	134	10000110	128	10000000
17	P9	B	2	00	153	10011001	152	10011000
18	P4	R	1	1	93	01011011	93	01011011

Total 40 bit pesan tersisip hasil pd Catatan ke 18.

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Langkah 6 : Load capacity
pixel still work terdapat = 10 dari 25 pixel total

$$\text{Load Capacity} = \frac{10}{25} \times 100\% = 72\%$$

Langkah 7 : perse PSNR

$$MSE = \frac{1}{N} \sum (\text{cover} - \text{stego})^2 = \frac{70}{75} = 1.04$$

$N = 25 \text{ pixel} \times 3 \text{ channel} = 75 \text{ nilai}$

total Substansi Kuadrat = 70

$$PSNR = 10 \log_{10} \left(\frac{255^2}{MSE} \right) = 10 \log_{10} \left(\frac{65025}{1.04} \right) = 47.96 \text{ dB}$$

PSNR > 40 dB, menunjukkan kualitas citra stego sangat baik.